

Grade 8
Unit 14
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

As part of a game, a fair six-sided die is rolled and a marble is randomly chosen from a bag containing one red marble, one blue marble, one green marble, and one orange marble. A representation of the marbles is shown.

1. Create a list, table, or tree diagram to determine the sample space.



		Pick 2			
Roll 1		R	B	G	O
	1	(1, R)	(1, B)	(1, G)	(1, O)
	2	(2, R)	(2, B)	(2, G)	(2, O)
	3	(3, R)	(3, B)	(3, G)	(3, O)
	4	(4, R)	(4, B)	(4, G)	(4, O)
	5	(5, R)	(5, B)	(5, G)	(5, O)
	6	(6, R)	(6, B)	(6, G)	(6, O)

2. Find the theoretical probability of rolling a 2 and the orange marble is randomly chosen.

$$P(2, \text{Orange}) = \frac{1}{24}$$

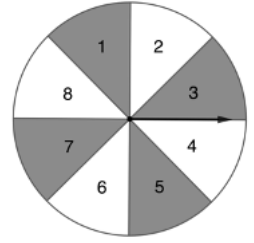
3. Find the theoretical probability that the die *does not* land on a 3 and the marble randomly chosen is red.

$$P(\text{not } 3, \text{red}) = \frac{5}{24}$$

4. What is the theoretical probability that the die lands on an odd number and the marble randomly chosen is a primary color (red or blue)?

$$P(\text{odd number, primary color}) = \frac{6}{24} \text{ or } \frac{1}{4}$$

In a two-step experiment a fair spinner is spun and a card is randomly picked from a standard deck of cards with 52 cards in four suits - hearts, diamonds, spades, and clubs. The spinner is shown.



5. What is the theoretical probability of landing on 5 on the fair spinner, $P(5)$?

$$\frac{1}{8}$$

6. What is the theoretical probability of picking a card with hearts from the deck of cards, $P(\text{hearts})$?

$$\frac{13}{52} = \frac{1}{4}$$

7. What is the theoretical probability of landing on 5 then picking a card with hearts from the deck, $P(5, \text{hearts})$?

$$\frac{1}{8} \times \frac{1}{4} = \frac{1}{32}$$

In a repeated experiment, a card is randomly chosen from a standard deck of cards twice, with replacement.

8. What is the probability of picking a diamond for the first pick, $P(\text{diamond})$?

$$\frac{13}{52} = \frac{1}{4}$$

9. What is the probability of picking a diamond both times, with replacement, $P(\text{all diamonds})$?

$$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

10. What is the probability of picking a spade both times, with replacement, $P(\text{all spades})$?

$$P(\text{spades}) \frac{13}{52} = \frac{1}{4}$$

$$P(\text{all spades}) \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$